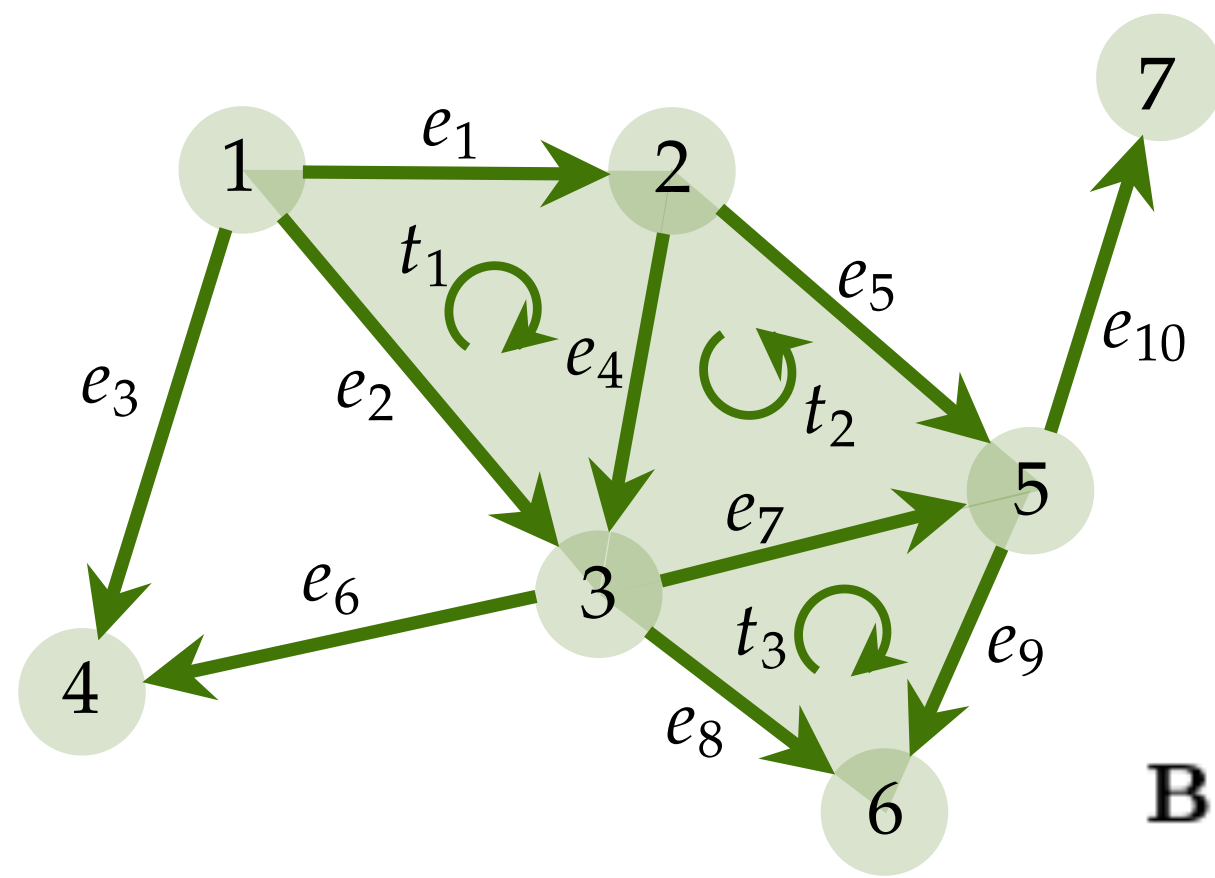


Representations of simplicial 2-complex

Incidences & Laplacians



Node-to-Edge

$$\mathbf{B}_1 = \begin{matrix} & \begin{matrix} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 & e_8 & e_9 & e_{10} \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \end{matrix} & \begin{pmatrix} -1 & -1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & -1 & -1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \end{matrix},$$

Edge-to-Faces

$$\mathbf{B}_2 = \begin{matrix} & \begin{matrix} t_1 & t_2 & t_3 \end{matrix} \\ \begin{matrix} e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \\ e_6 \\ e_7 \\ e_8 \\ e_9 \\ e_{10} \end{matrix} & \begin{pmatrix} 1 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \end{matrix}$$

Graph Laplacian: $\mathbf{L}_0 = \mathbf{B}_1 \mathbf{B}_1^\top = D - A$

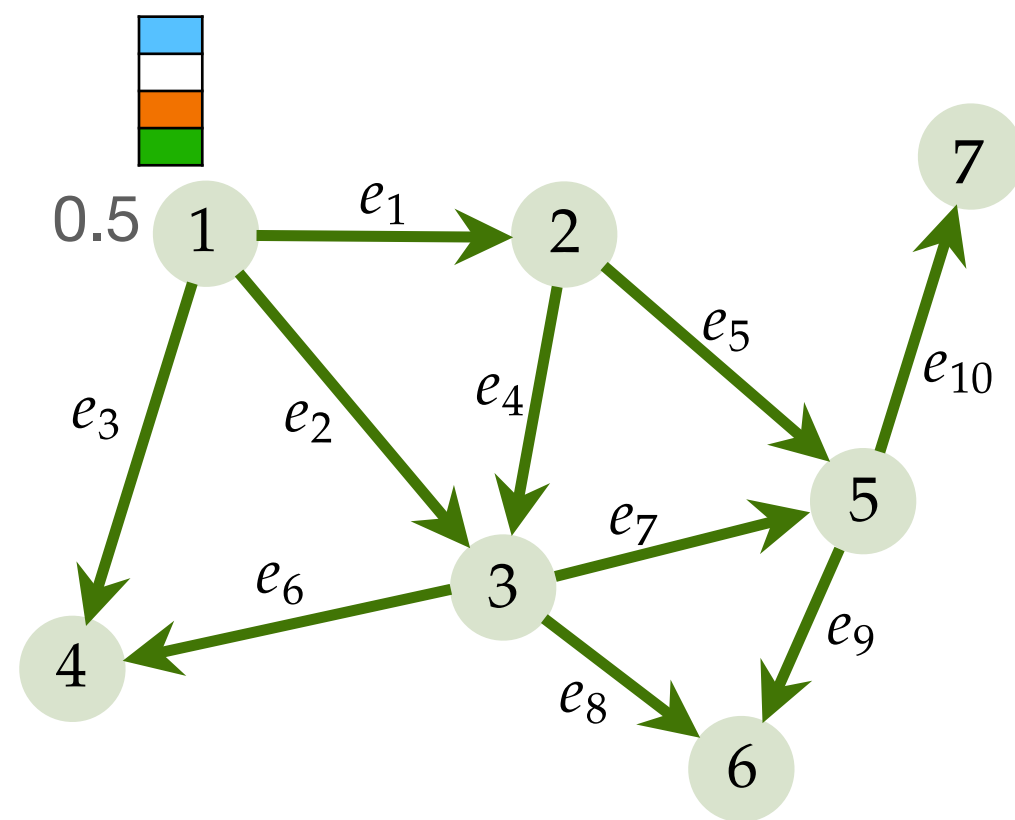
1-Hodge Laplacian: $\mathbf{L}_1 = \mathbf{B}_1^\top \mathbf{B}_1 + \mathbf{B}_2 \mathbf{B}_2^\top := \mathbf{L}_{1,d} + \mathbf{L}_{1,u}$

Up adjacency via faces

Down adjacency
via nodes

Data on simplices

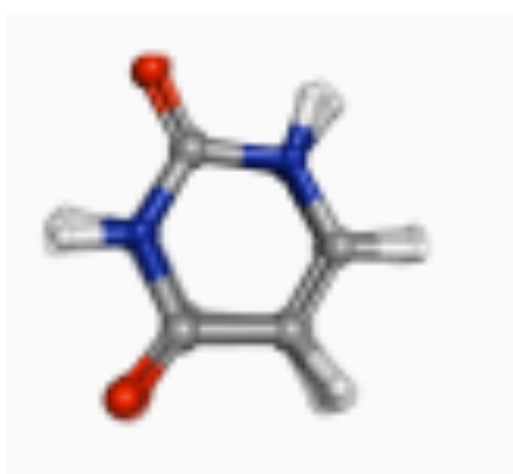
Signals on nodes, edges, triangles, ...



Node function

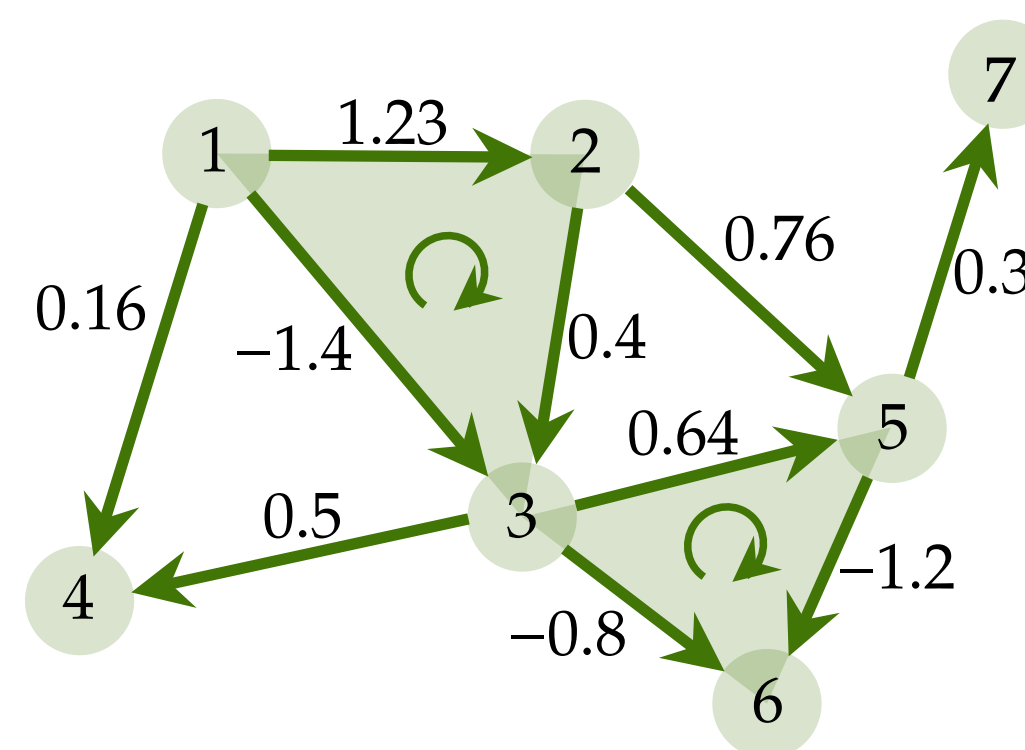
$$f_0 : V \rightarrow \mathbb{R}$$

$$\mathbf{f}_0 = (f_0(1), \dots, f_0(N_0))^T$$



Atom coordinates

$$f_0 : V \rightarrow \mathbb{R}^3$$



Edge function

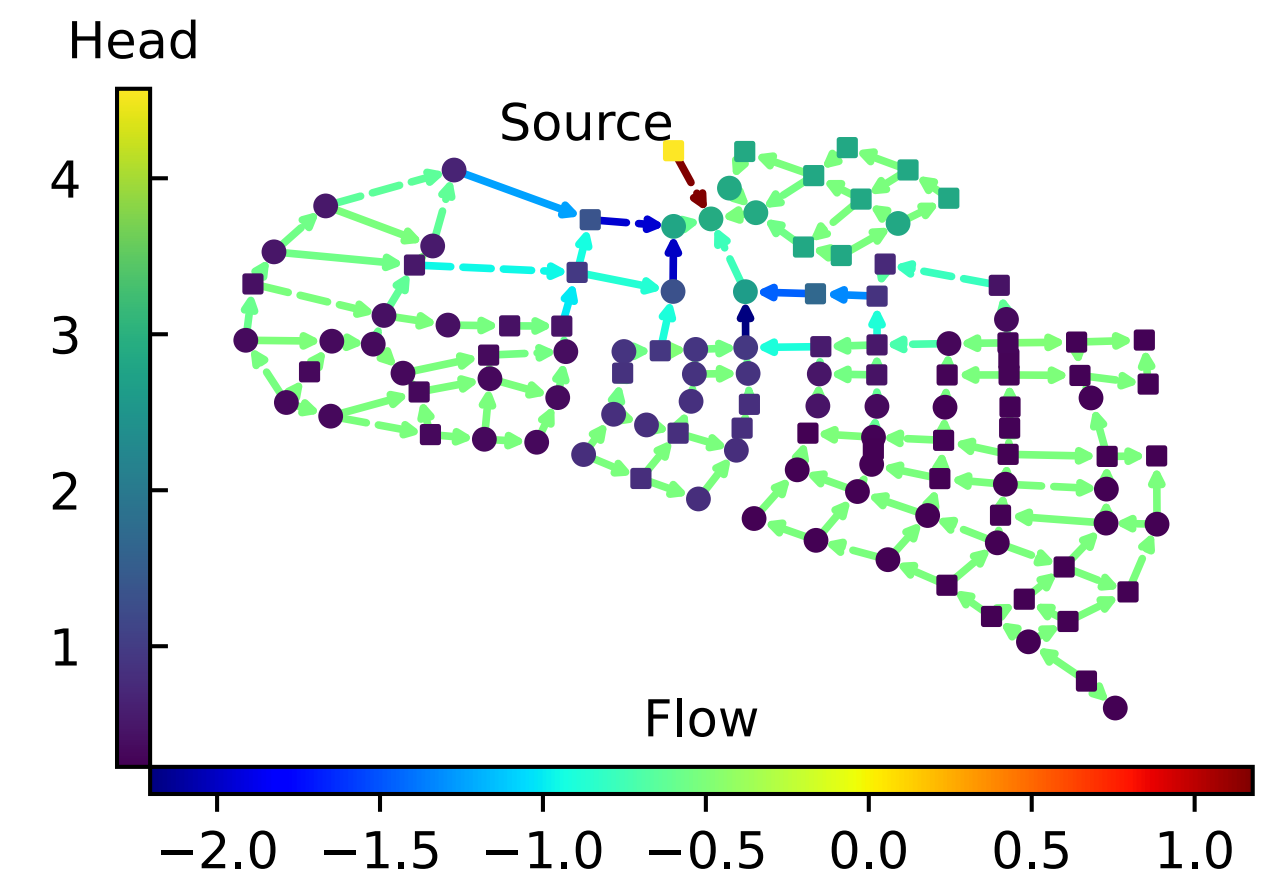
$$f_1 : E \rightarrow \mathbb{R}$$

$$\mathbf{f}_1 = (f_1(e_1), \dots, f_1(e_{N_1}))^T$$

- Alternating property

- Magnitude and sign

- Flow-type data (natural)
 - traffic/water flow, **information flow** in gene regulation...
- Edge-based vector field discretisation (computational)
- ...
- Representation learning
 - High-dim edge features
- Geometric algebra transformer 2023



Water flows (Yang et al. 2023)